MAMInet II: Illumination-Insensitive Modalitywise Assimilation Guided Multistage Interaction Network for Hyperspectral and LiDAR Joint Classification

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***Abstract*— Fusing hyperspectral (HS) and light detection and ranging (LiDAR) data is capable of enhancing land-cover interpretation capability in multimodal remote sensing (RS) tasks. Nevertheless, nonuniform surrounding illumination in real-world capturing scenarios tends to distort the spectral curves of on-ground objects, inevitably resulting in poor classification performance. This nonuniform illumination further interferes with the cross-modal RS information interaction procedure. In this article, an Illumination Insensitive Modalitywise Assimilation guided Multistage Interaction network (MAMInet II) is proposed to tackle this challenge. Our primary motivation is to eliminate complicated illumination interference (CII) and capture high-level modalitywise assimilation information (MAI) from HS and LiDAR data simultaneously. Notably, the geological connection between these two RS modalities is emphasized in MAMInet II, making the joint classification framework interpretable. Specifically, MAMInet II presents one channelwise illumination removal mechanism by simulating real-world illumination conditions and associating HS spectral curves with LiDAR elevations. One generative switching mechanism contributes to closing the cross-modal RS distribution gap and enhancing the cross-modal capturing ability. These mechanisms are extended to multiscale and multistage versions for aligning local and global RS properties. Experiments illustrate that MAMInet II outperforms several state-of-the-art techniques qualitatively and quantitatively. All source codes are available at https://github.com/weixintttt/MAMInet-II.**

***Index Terms*— Classification, generative model, hyperspectral (HS), light detection and ranging (LiDAR), nonuniform illumination.**

I. Introduction

REMOTE sensing (RS) techniques are widely utilized in capturing on-ground object descriptions and patterns, e.g., resource estimation, atmospheric observation, and agricultural protection [1]. Various types of satellites and aircraft are applied for capturing RS modalities, including hyperspectral (HS) data [2] and light detection and ranging (LiDAR) data [3]. Specifically, HS data contain spatial and spectral information of on-ground objects through their abundant but redundant spectral channels. As a complementary RS information source, LiDAR data record elevations by actively transmitting and receiving pulsed laser lights. Existing studies have verified that fusing HS and LiDAR data for joint classification fully leverages the advantages of multisource RS detection. Notably, digital surface models (DSMs) derived from LiDAR data are always considered in related works considering their capabilities of describing terrain and artificial surfaces.

Diverse deep learning (DL) architectures for joint classification have been presented in real-world tasks [4]. Classic techniques are generally divided into feature-level fusion and decision-level fusion [5]. Feature-level fusion extracts discriminant information from diverse perspectives for subsequent fusion processes. Yu et al. [6] presented an HS intrinsic image decomposition (HIID) guided data fusion network called HI^{2}D^{2}FNet from the intrinsic perspective. Li et al. [7] proposed a spatial–spectral saliency reinforcement network (Sal^{2}RN) for accurate classification results. To simulate multiview latent spaces, Feng et al. [8] presented one joint classification framework based on multiview feature learning and multilevel fusion. Zhang et al. [9] designed an information fusion network called interleaving perception convolutional neural network (IP-CNN). Decision-level fusion with multiple voting classifiers tends to be connected with feature-level fusion for superior classification performance [10], [11], [12].

These techniques illustrate the effectiveness of DL networks in joint classification tasks. Generally, existing DL structures are divided into discriminative and generative models [13], [14], [15]. Discriminative models simulate conditional probabilities and learn category boundaries by stacked layers. Yu et al. [16] proposed one spatial–spectral stacked convolutional neural network (CNN) with one feedback attention mechanism. Yang et al. [17] and Sun et al. [18] considered hybrid mechanisms for satisfactory classification performance. Despite their excellent performance, discriminative models encounter challenges when dealing with limited samples and unbalanced categories since inherent RS characteristics are hard to acquire. In contrast, generative models simulate joint RS probability distributions and classified categories,

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attempting to capture underlying generation procedures rather than directly optimizing for great accuracies [19]. Typically, as an autoencoder (AE) variant [20], variational AE (VAE) learns underlying distributions and appends constraints on its encoder to restrain the latent distribution [21], [22], [23]. Its strong generative ability enhances the potential of information mining. Yu et al. [24] considered VAE-based classification modules and regarded mean vectors as pure latent features to improve robustness.

Existing multimodal fusion approaches achieve noteworthy performance when aligning RS modalities. Nevertheless, exploring excellent and effective fusion strategies for HS and LiDAR data is hampered by two challenges.

1) *Illumination Interference:* As one typical passive optical RS modality, HS data are particularly susceptible to nonuniform surrounding illumination in real-world scenarios. Specifically, spectral curves of objects will be severely distorted, leading to poor classification performance. Separate illumination removal always fails to utilize comprehensive land-cover descriptions. This step inevitably results in information loss and optimization inconsistency with subsequent joint classification procedures.
2) *Heterogeneous Simulation:* High modality consistency is required by multimodal fusion approaches in terms of information alignment and exchange. Considering the diverse capturing instruments and theories, HS and LiDAR data contain heterogeneous descriptions, significantly impeding joint classification accuracies and efficiencies. Thus, accurately simulating heterogeneous RS modalities and capturing their inherent connection are primary tasks.

To solve these two challenges, an Illumination-Insensitive Modalitywise Assimilation guided Multistage Interaction network called MAMInet II is proposed. MAMInet II consists of three modules, including the latent space simulation (LSS) module, the insensitive illumination removal (IIR) module, and the generative distribution switching (GDS) module. The core motivation, provided in Fig. 1, is to capture high-level modalitywise assimilation information (MAI) from HS and LiDAR data simultaneously for joint classification in real-world scenarios with nonuniform surrounding illumination.

1) For the first challenge, MAMInet II simulates and enhances the geological connection between HS and LiDAR data to eliminate complicated illumination interference (CII), making the fusion process interpretable. It shows that, guided by LiDAR data, MAMInet II designs one channelwise illumination removal mechanism to decompose HS data into one reflectance component with object material information and one shading component with obtrusive illumination information. Compared with the latter, the former is more highly relevant to the annotated ground truth. In terms of HS and LiDAR properties, the decomposition procedure is constrained by connecting illumination with detailed elevations to eliminate CII.
2) For the second challenge, MAMInet II simulates this heterogeneity and closes the distribution gap. Motivated by classic VAE, MAMInet II learns high-level MAI and presents a unique generative switching mechanism for aligning HS and LiDAR data. The VAE motivated structure significantly increases the information mining potential and facilitates cross-modal interaction. The multiscale and multistage versions further align local and global RS properties. The main contributions are summarized as follows.

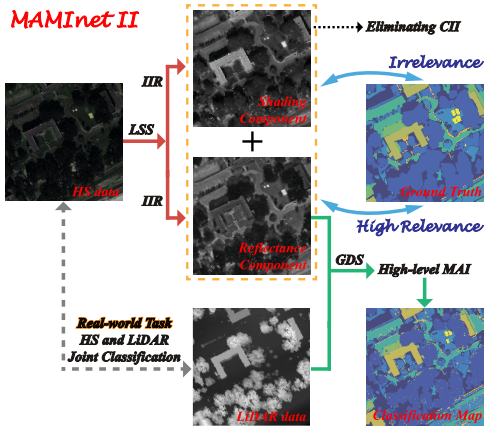


Fig. 1. Core motivation of MAMInet II for an intuitive explanation.

1) A novel HS and LiDAR joint classification method called MAMInet II is proposed to capture high-level MAI from both modalities simultaneously in real-world scenarios with nonuniform surrounding illumination. To the best of our knowledge, it is the first work that combines interpretable illumination removal and generative switching mechanisms to enhance the cross-modal information mining capability in complicated RS acquisition scenarios.
2) MAMInet II eliminates CII during the whole joint classification procedure by simulating and emphasizing the geological connection between these two specific RS modalities. Its channelwise illumination removal mechanism associates HS spectral curves with LiDAR elevations in terms of their properties. This structure significantly increases the fusion capability and interpretability.
3) MAMInet II closes the cross-modal RS distribution gap and enhances the cross-modal capturing ability by presenting one VAE motivated generative switching mechanism. This mechanism contributes to increasing the potential of information mining and providing opportunities for cross-modal interaction. Furthermore, MAMInet II extends these mechanisms into multiscale and multistage versions for further aligning local and global RS properties.

Section II provides some related works. Section III describes MAMInet II in detail. Section IV provides and analyzes the experimental results. Section V concludes this work.

II. Related Works

A. HS and LiDAR Fusion and Joint Classification

Fusing HS and LiDAR data for joint classification is beneficial for accurately identifying the detailed categories in numerous scenarios [25]. Diverse works capture significant approaches for enhancing the fusion performance of

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these two RS modalities [26]. Brell et al. [27] fused HS reflectance information and LiDAR distance measurement to increase data geometric accuracies. Zhang et al. [28] introduced one transformer-driven network for multimodal fusion to capture specific and shared RS properties simultaneously. Han et al. [29] presented a cross-modal semantic enhancement network for multimodal semantic information mining and fusion. Furthermore, several studies consider diverse attention-based modules to strengthen local and global characteristics [30], [31]. Despite their satisfactory performance, they tend to regard HS and LiDAR data as one ordinary RS modality pair, without considering their specific capturing properties. MAMInet II captures their complementary geometric descriptions in the spatial domain for increasing the fusion capability and interpretability.

B. HS Intrinsic Image Decomposition

MAMInet II removes the effect of nonuniform surrounding illumination on HS and LiDAR fusion performance. HIID performs well in shadow removal and image enhancement. Generally, HIID decomposes HS data into one reflectance component $\boldsymbol{R}$ and one shading component $\boldsymbol{S}$. Different from existing IID approaches for RGB data, the components in HIID share the same channel number with the HS data for distinctly describing spectral characteristics. Formally, the HS data are denoted as $\boldsymbol{H} \in \mathbb{R}^{a\times b\times c}$, where a and b are the height and width, and c is the channel number. The HIID is represented as

$$\boldsymbol{H}(\lambda) = \boldsymbol{R}(\lambda) \odot \boldsymbol{S}(\lambda), \quad 1 \le \lambda \le c \tag{1}$$

where $\odot$ is the Hadamard product. Notably, $\boldsymbol{R}$ describes the object materials in uniform surrounding illumination and $\boldsymbol{S}$ reflects the detailed illumination. For the targets from the same category in nonuniform illumination, their spectral curves can be revised by removing the decomposed $\boldsymbol{S}$ and retaining the remaining $\boldsymbol{R}$. To optimize the ill-posed formula in (1), multiple constraints on components have been presented [32], [33]. Gao et al. [34] extended HIID application scopes by designing temporal-spatial constraints for change detection. In this article, the decomposition mechanism of HIID is simulated in MAMInet II by highlighting the specific geological connection between HS and LiDAR data. The illumination removal procedure is integrated into MAMInet II rather than regarded as one separate preprocessing step. The reason is that this integration contributes to unifying optimization directions.

C. Variational Autoencoder

MAMInet II captures high-level MAI for closing the cross-modal distribution gap and enhancing the cross-modal capturing ability. As an extension of AE, VAE provides the potential for RS information switching. It considers random Gaussian noise when reconstructing its input, making the trained network robust regarding noise interference. For one input $\boldsymbol{x}$, VAE makes the output have satisfactory similarity with $\boldsymbol{x}$. The output generated from the conditional distribution $P(\boldsymbol{x}|\boldsymbol{z})$ is reconstructed from a hidden variable $\boldsymbol{z}$ generated from the prior distribution $P(\boldsymbol{z})$. Considering the posterior distribution $P(\boldsymbol{z}|\boldsymbol{x})$ cannot be obtained, it is replaced by a Gaussian distribution $Q(\boldsymbol{z}|\boldsymbol{x})$ for convenience. The evidence lower bound (ELBO) function can be formulated as

$$\text{ELBO} = \mathbb{E}_{Q(\boldsymbol{z}|\boldsymbol{x})} P(\boldsymbol{x}|\boldsymbol{z}) - \text{KL}[P(\boldsymbol{z}|\boldsymbol{x}), Q(\boldsymbol{z}|\boldsymbol{x})] \tag{2}$$

where $\mathbb{E}_{Q(\boldsymbol{z}|\boldsymbol{x})} P(\boldsymbol{x}|\boldsymbol{z})$ calculates the distance between $\boldsymbol{x}$ and the output by mean squared error, $\text{KL}[P(\boldsymbol{z}|\boldsymbol{x}), Q(\boldsymbol{z}|\boldsymbol{x})]$ measures the distance between $P(\boldsymbol{z}|\boldsymbol{x})$ and $Q(\boldsymbol{z}|\boldsymbol{x})$ by Kullback–Leibler (KL) divergence. Considering $Q(\boldsymbol{z}|\boldsymbol{x})$ follows a Gaussian distribution, its mean vector and standard deviation vector are denoted as $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$, respectively. Kingma and Welling [35] regarded the intractable $P(\boldsymbol{z}|\boldsymbol{x})$ as $\mathcal{N}(0, 1)$. Thus, the second element in (2) can be rewritten as

$$\begin{aligned}\text{KL}[P(\boldsymbol{z}|\boldsymbol{x}), Q(\boldsymbol{z}|\boldsymbol{x})] &= \text{KL}\left[\mathcal{N}(0, 1), \mathcal{N}\left(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\right)\right] \\ &= \left(\boldsymbol{\mu}^2 + \boldsymbol{\sigma}^2 - \log \boldsymbol{\sigma}^2 - 1\right)/2\end{aligned} \tag{3}$$

where $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2) = \boldsymbol{\mu} + \boldsymbol{\sigma} \times \mathcal{N}(0, 1)$. VAE captures the specific distribution pattern in latent spaces, providing the potential for information switching. Furthermore, the robustness of VAE can be enhanced by introducing randomly sampling vectors continuously. The mean vector $\boldsymbol{\mu}$ and standard deviation vector $\boldsymbol{\sigma}$ quantitatively constrain the dynamic training procedure. In this article, MAMInet II leverages these strengths in HS and LiDAR joint classification tasks for satisfactory performance.

III. Methodology

The scheme of MAMInet II with three modules is shown in Fig. 2. HS data and one lightweight LiDAR modality, i.e., DSM, are chosen for joint classification.

A. LSS Module

The LSS module simulates the latent spaces in the HS and LiDAR data to extract features with high quality. Formally, the HS and LiDAR data are represented as $\boldsymbol{H} \in \mathbb{R}^{a\times b\times c}$ and $\boldsymbol{L} \in \mathbb{R}^{a\times b}$, respectively, where a and b are the length and width, and c is the HS channel number. To capture spatial information, a $w \times w$ sliding window is considered to partition $\boldsymbol{H}$ and $\boldsymbol{L}$ into small patches $\boldsymbol{H_i} \in \mathbb{R}^{w\times w\times c}$ and $\boldsymbol{L_i} \in \mathbb{R}^{w\times w}$, where $1 \le i \le a \times b$. Notably, w corresponds to the capturing scope of spatial information. MAMInet II aligns local and global RS properties for accurate joint classification by the multiscale and multistage versions, which are fully analyzed in Section III-D. Thus, the following structures are designed for the scale w. In the HS-related stream, two 2-D convolution (Conv2d) layers, i.e., C_1 and C_2 with c 3×3 kernels (use SAME padding), contribute to capturing spatial information from two randomly selected patches $\boldsymbol{H_x}$ and $\boldsymbol{H_y}$. Their inputs are denoted as $\boldsymbol{h_1}$ and $\boldsymbol{h_2}$. The reason why two HS patches are utilized is that MAMInet II simulates neighborhood and nonneighborhood relationships. Similarly, C_3 and C_4 whose inputs are $\boldsymbol{l_1}$ and $\boldsymbol{l_2}$ capture spatial information from $\boldsymbol{L_x}$ and $\boldsymbol{L_y}$. Notably, $\boldsymbol{H_x}$ corresponds to $\boldsymbol{L_x}$. The outputs of these Conv2d layers are activated by the Tanh function and normalized by batch normalization (BN) layers. The shapes of the LSS module outputs, i.e., $\overline{\boldsymbol{h}}_1$, $\overline{\boldsymbol{h}}_2$, $\overline{\boldsymbol{l}}_1$, and $\overline{\boldsymbol{l}}_2$, are $[bs, w, w, c]$, where bs is the batch size. This module captures multimodal features and provides convenience for the decomposition procedures.

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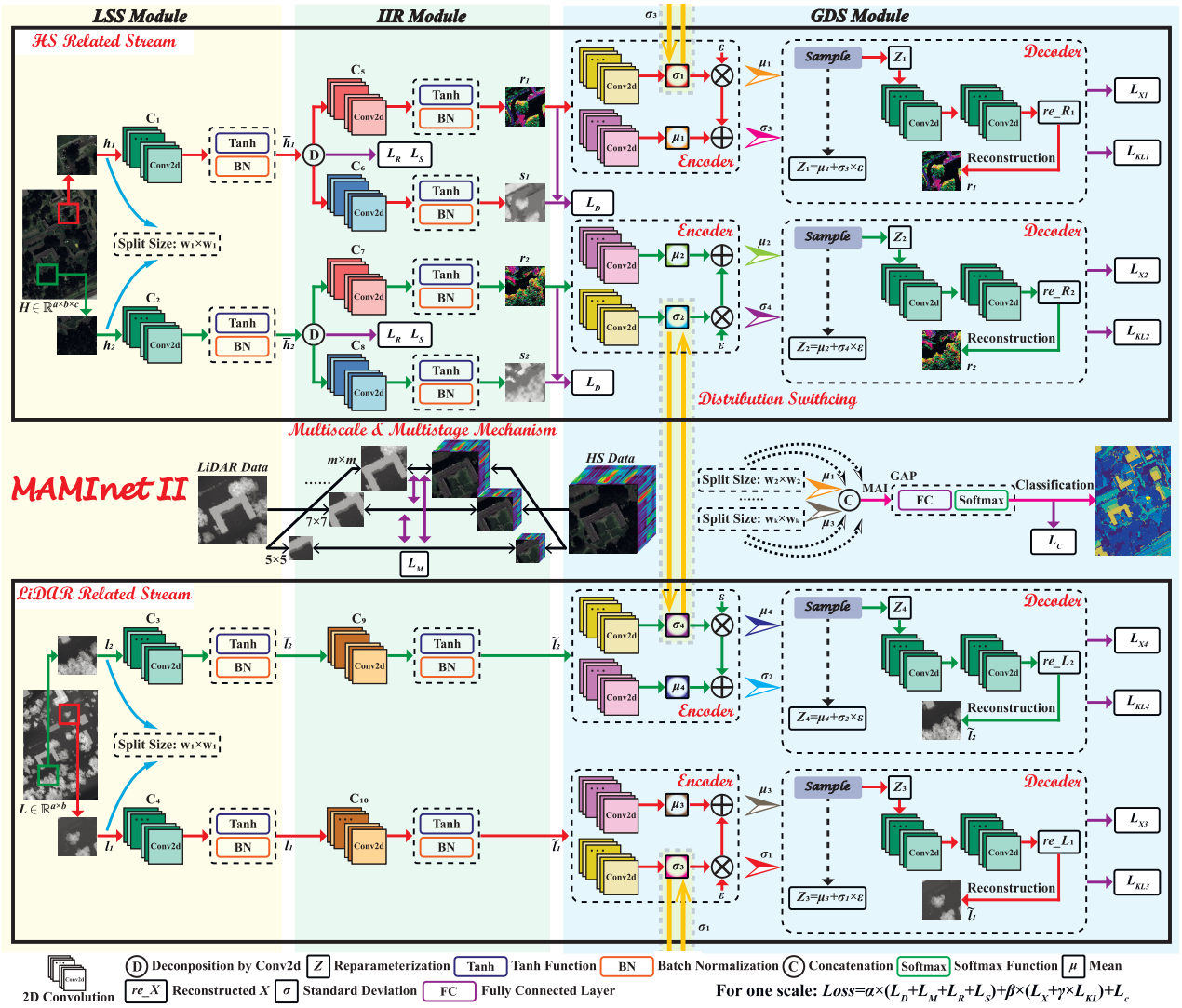


Fig. 2. Scheme of MAMInet II for HS and LiDAR joint classification The network consists of the LSS module, the IIR module, and the GDS module. In the LSS module, two sample pairs are randomly selected for simulating RS samples in neighboring regions. The IIR module eliminates CII by constraining the decomposition procedure in terms of specific RS properties and connections. The GDS module contains several encoders and decoders for closing the distribution gap. The $\boldsymbol{\mu}$s and $\boldsymbol{\sigma}$s are marked in different colors. Based on the functions, all arrows and Conv2d layers are marked in different colors.

B. IIR Module

The IIR module with one channelwise illumination removal mechanism contributes to eliminating CII by simulating and enhancing the geological connection between HS and LiDAR data. It yields satisfactory performance in nonuniform surrounding illumination, making the whole framework interpretable. Inspired by HIID, the IIR module attempts to decompose the HS data into one reflectance component, describing object materials, and one shading component, reflecting detailed surrounding illumination. The whole decomposition procedure is guided by the LiDAR data. Inspired by several existing works, MAMInet II assumes that neighboring objects with similar elevations share similar surrounding illumination [6], [36]. Notably, MAMInet II does not assume that objects with larger elevations correspond to brighter illumination. Although surrounding illumination can be influenced by several factors, including solar elevation angle, capturing time, and occlusion, these factors are not considered in this work due to the following considerations: 1) for a single HS dataset covering a limited area, the solar elevation angles and capturing times for all samples are consistent and negligible and 2) for an HS dataset with relatively low spatial resolution, occlusion always exist around on-ground objects and occupies a limited segment of the overall data. Moreover, this assumption contributes to generating reasonable components and can be revised by the following optimization procedure in terms of its roughness. Based on (1), both components can be reconstructed as

$$L_D = \frac{1}{c}\sum_{i,j}[\mathrm{GAP}(\boldsymbol{h_i} - \boldsymbol{r_i} \odot \boldsymbol{s_i})]_j^2 \tag{4}$$

where GAP$(\cdot)$ is the global average pooling (GAP) layer, and c is the HS channel number, $i \in \{1, 2\}$, $1 \le j \le c$. In Fig. 2, the red and blue Conv2d layers belong to the reflectance and shading blocks, respectively. For the reflectance block, the input shapes of C_5 and C_7 are $[bs, w, w, c]$. Their parameters are consistent with those of C_1 and C_2. This block captures high-precision reflection components for the reconstruction in

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the GDS module. Subsequently, the Tanh function and BN layers are utilized. To increase the quality of the decomposed reflectance component, a constraint based on HS properties is introduced. Specifically, MAMInet II assumes that adjacent spectral curves in the spatial domain are highly relevant to their detailed materials and Euclidean distance. Thus, MAMInet II defines the HS gradient $\boldsymbol{g}^{\mathrm{HS}} = \{\boldsymbol{g}_1^{\mathrm{HS}}, \boldsymbol{g}_2^{\mathrm{HS}}, \ldots, \boldsymbol{g}_c^{\mathrm{HS}}\}$ for decomposition in the IIR module, which is formulated as

$$\begin{cases} \boldsymbol{g}_j{}^{\mathrm{HS}} = \mathrm{Sigmoid}\left\{[|\Delta\mathrm{GAP}(\boldsymbol{h})| \cdot \mathrm{Euc}(\boldsymbol{h}_1, \boldsymbol{h}_2)]^{-1}\right\} \\ |\Delta\mathrm{GAP}(\boldsymbol{h})| = |\mathrm{GAP}(\boldsymbol{h}_1) - \mathrm{GAP}(\boldsymbol{h}_2)|_j \end{cases} \tag{5}$$

where Sigmoid is the Sigmoid function, $\mathrm{Euc}(\cdot)$ is the Euclidean distance, and $|\cdot|$ is the absolute value operator. Therefore, the reflectance component is constrained as

$$L_R = \frac{1}{c} \sum_{1 \le j \le c} \left\{ \boldsymbol{g}_j{}^{\mathrm{HS}} [\mathrm{GAP}(\boldsymbol{r}_1) - \mathrm{GAP}(\boldsymbol{r}_2)]_j^2 \right\} \tag{6}$$

where the reflectance features $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ correspond to $\boldsymbol{h}_1$ and $\boldsymbol{h}_2$, respectively. The shading block, aiming to capture high-precision shading components, shares a similar structure with the reflectance block. It learns spatial information by C_6 and C_8, which have the same parameters with C_5 and C_7. Apparently, the construction motivations of both blocks are different in terms of their detailed functions. Compared with the reflectance block, the shading block attempts to simulate surrounding illumination and environment to eliminate CII. The shading component is constrained by highlighting the geological connection between HS and LiDAR data. Thus, the detailed constraint is provided after analyzing the LiDAR block. The LiDAR block in the LiDAR-related stream corresponds to the orange Conv2d layers. Its C_9 and C_{10} capture prominent elevation features, making both streams consistent. The outputs of this block are $\tilde{\boldsymbol{l}}_1$ and $\tilde{\boldsymbol{l}}_2$ for $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$, respectively. Both the shading component and the LiDAR data are not related to categories, providing inspiration for the constraint on the shading block. Specifically, the on-ground objects with similar elevations tend to have consistent environmental illumination. MAMInet II captures the geological connection between HS and LiDAR data by considering the detailed elevations for generating the shading component in the IIR procedure. The detailed constraint is formulated as

$$\begin{cases} \boldsymbol{g}_j{}^{L} = \mathrm{Sigmoid}\left\{[|\Delta\mathrm{GAP}(\boldsymbol{l})| \cdot \mathrm{Euc}(\boldsymbol{l}_1, \boldsymbol{l}_2)]^{-1}\right\} \\ |\Delta\mathrm{GAP}(\boldsymbol{l})| = |\mathrm{GAP}(\boldsymbol{l}_1) - \mathrm{GAP}(\boldsymbol{l}_2)|_j \end{cases} \tag{7}$$

where $\boldsymbol{g}^{\boldsymbol{L}} = \{\boldsymbol{g}_1^{\boldsymbol{L}}, \boldsymbol{g}_2^{\boldsymbol{L}}, \ldots, \boldsymbol{g}_c^{\boldsymbol{L}}\}$ is the LiDAR gradient. The LiDAR gradient contributes to simulating the correlation between neighborhoods and nonneighborhoods. When $\boldsymbol{g}^{\boldsymbol{L}}$ between two samples are small, the corresponding spatial distance and elevation difference should also be small, meaning that their corresponding shading samples should be consistent. Thus, the shading component is constrained as

$$L_S = \frac{1}{c} \sum_{1 \le j \le c} \left\{ \boldsymbol{g}_j{}^{L} \left[\mathrm{GAP}(\boldsymbol{s}_1)_j - \mathrm{GAP}(\boldsymbol{s}_2)_j \right]^2 \right\}. \tag{8}$$

By introducing $\boldsymbol{g}^{\mathrm{HS}}$ and $\boldsymbol{g}^{\boldsymbol{L}}$, the decomposition procedure is connected with these two specific RS modalities. Inspired by this goal, the reflectance and shading components are highly connected with HS spectral curves and LiDAR elevations to eliminate CII in nonuniform surrounding illumination.

C. GDS Module

The GDS module with one generative switching mechanism contributes to closing the cross-modal RS distribution gap and enhancing the cross-modal capturing ability. Its generative switching mechanism is motivated by VAE to increase the potential of information mining and provide opportunities for cross-modal interaction. VAE reconstructs its input and decomposes the input distribution into one mean vector and one standard deviation vector. Specifically, with the contributions of several Conv2d layers (the yellow and purple Conv2d layers correspond to the mean vector and standard deviation vector), the generated $\boldsymbol{r}_1$ from $\boldsymbol{h}_1$ can be decomposed into $\boldsymbol{\mu}_1$ and $\boldsymbol{\sigma}_1$. To remove the CII effect on joint classification, the generated $\boldsymbol{s}_1$ is neglected for enhancing category consistency and eliminating CII. Following this strategy, the HS reflectance features $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ and the LiDAR features $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$ can be decomposed into the mean vectors $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\mu}_3, \boldsymbol{\mu}_4\}$ and the standard deviation vectors $\{\boldsymbol{\sigma}_1, \boldsymbol{\sigma}_2, \boldsymbol{\sigma}_3, \boldsymbol{\sigma}_4\}$. To close the cross-modal RS distribution gap for high-level MAI, the HS and LiDAR standard deviations for the same patch are switched (i.e., $\boldsymbol{\sigma}_1$ and $\boldsymbol{\sigma}_3$, $\boldsymbol{\sigma}_2$ and $\boldsymbol{\sigma}_4$) since abundant distribution properties and patterns can be gathered in standard deviations by VAE. These switching vectors are considered in the following sampling procedure by introducing random noise $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, 1)$ as $\boldsymbol{Z}_{\boldsymbol{p}} = \boldsymbol{\mu}_{\boldsymbol{p}} + \boldsymbol{\sigma}_{\boldsymbol{q}} \times \boldsymbol{\varepsilon}$, where $\boldsymbol{Z}_{\boldsymbol{p}}$ is the reparameterization feature, $[p, q] \in \{[1, 3], [2, 4], [3, 1], [4, 2]\}$. Subsequently, these $\boldsymbol{Z}_{\boldsymbol{p}}$s are used for reconstruction in the decoders. The decoders with two Conv2d layers aim to increase the similarities between their outputs and corresponding patches. When these similarities are quite high, the captured mean vectors are capable of simulating the category relationship for satisfactory performance. Considering that $\boldsymbol{\mu}_1 \& \boldsymbol{\mu}_3$ and $\boldsymbol{\mu}_2 \& \boldsymbol{\mu}_4$ are for two randomly selected patches, MAMInet II chooses the first patch for classification and captures the MAI feature as $\boldsymbol{M} = \mathrm{Concat}(\boldsymbol{\mu}_1, \boldsymbol{\mu}_3) \in \mathbb{R}^{w \times w \times 2c}$, where Concat is the concatenation operator. Then, the MAI feature is classified by one FC layer and activated by the Softmax function. Notably, the GAP layers change the feature shape from $\mathbb{R}^{w \times w \times 2c}$ to $\mathbb{R}^{1 \times 1 \times 2c}$. Inspired by (3), the distribution distance constraint in the GDS module is formulated as

$$L_{\mathrm{KL}} = \frac{1}{c} \sum_{1 \le j \le c} \left\{ \sum_{1 \le i \le 4} \left[\left(\boldsymbol{\mu}_i{}^2 + \boldsymbol{\sigma}_i{}^2 - \log \boldsymbol{\sigma}_i{}^2 - 1 \right)/2 \right]_j \right\}. \tag{9}$$

Besides the distribution distance, the input and output of the AE variant should be similar, which is constrained as

$$L_X = \frac{1}{c} \sum_{1 \le j \le c} \left\{ \sum_{1 \le i \le 2} \boldsymbol{\Phi}_{i,j} \right\} \tag{10}$$

where $\|\cdot\|_2$ represents the l_2 norm, $\boldsymbol{\Phi}_{i,j} = \| (\boldsymbol{re_R}_i - \boldsymbol{r}_i)_j \|_2^2 + \| (\boldsymbol{re_L}_i - \tilde{\boldsymbol{l}}_i)_j \|_2^2$, $\boldsymbol{re_R}_i$, and $\boldsymbol{re_L}_i$ are the reconstructed reflectance components and LiDAR features, respectively, $\boldsymbol{r}_i$ and $\tilde{\boldsymbol{l}}_i$ are the inputs of the GDS module. The cross-entropy-based constraint for classification is formulated as

$$L_C = \sum_{1 \le j \le l} -\boldsymbol{o}_j \log(\boldsymbol{out}_j) - (1 - \boldsymbol{o}_j) \log(1 - \boldsymbol{out}_j) \tag{11}$$

where $\boldsymbol{out}_j$ is the predicted label, $\boldsymbol{o}_j$ is the one-hot label of $\boldsymbol{out}_j$, and l is the category number.

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D. Multiscale and Multistage Version

MAMInet II further extends these mechanisms into multiscale and multistage versions to align local and global RS properties. First, these structures are specifically presented for the scale w, which is determined by the sliding window shape. To capture comprehensive knowledge, MAMInet II considers multiple scales $\{w_1, w_2, \ldots, w_k\}$ for generating MAI features $\{\boldsymbol{M_1}, \boldsymbol{M_2}, \ldots, \boldsymbol{M_k}\}$ from diverse perspectives, where k is the scale number. The structure for one single scale is divided into multiple stages, including the LSS, IIR and GDS modules, to make the whole network shipshape and convenient. These stages have diverse expectations: 1) the LSS module extracts high-quality features to simulate the HS and LiDAR latent spaces; 2) the IIR module attempts to decompose the HS data into high-quality and interpretable components with the help of the LiDAR data for eliminating CII in nonuniform illumination; and 3) the GDS module aims to close the cross-modal RS distribution gap for satisfactory joint classification performance. Although their expectations are different, these stages contribute to updating MAMInet II toward one consistent direction. Furthermore, Fig. 2 shows that the layers of these modules in the RS and LiDAR-related streams are also consistent in terms of their stages. These multiscale and multistage versions provide convenience for aligning cross-modal RS properties and introducing cross-modal constraints. The generated MAI features are homologous since their corresponding patches share the same center. Thus, a KL divergence-based constraint increases this homology as

$$L_M = \frac{1}{2c} \sum_{1 \le j \le 2c} \sum_{1 \le m \le k} \sum_{1 \le n \le k} \mathrm{KL}[\boldsymbol{M_m}, \boldsymbol{M_n}]_j \tag{12}$$

where $\boldsymbol{M_m}$ and $\boldsymbol{M_n}$ are two different MAI features, $m \neq n$. This constraint is adaptive to the multiscale and multistage versions in terms of information completeness and consistency.

E. Optimization

MAMInet II has two expectations: 1) decomposing high-quality and interpretable HS components for eliminating CII in nonuniform illumination and 2) capturing high-level MAI from both RS modalities for satisfactory joint classification performance. Considering these expectations and the unique network structure, the loss function L is formulated as

$$\begin{aligned} L = {} & \alpha(L_D + L_M + L_R + L_S) \\ & + \beta \times (L_X + \gamma \times L_{\mathrm{KL}}) + L_C \end{aligned} \tag{13}$$

where α and β are the loss weights and γ is the KL balance which is set to 0.01. The contributions of these loss elements are clearly analyzed in the aforementioned descriptions. The computing complexity of MAMInet II is $\mathcal{O}(w^2c^2)$, where w and c are the sliding window side and HS channel number. It shows that the optimizing time is mainly determined by w and c. Furthermore, this time is also influenced by multiple factors, e.g., the training epochs, in real-world scenarios.

IV. Experimental Results and Analysis

A. Dataset Description

The Trento dataset was acquired over the south of Trento, Italy. It has one HS data and one LiDAR derived DSM data.

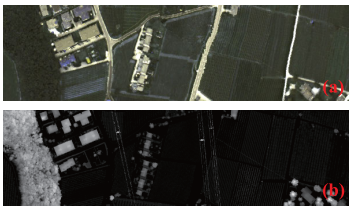


Fig. 3. Visualization of the Trento dataset, including (a) HS pseudo color image and (b) LiDAR DSM gray image.

TABLE I
Detailed Sample Numbers in the Trento Dataset

Class	Land-cover Category	Training	Testing	Total
1	Apple Trees	40	3994	4034
2	Buildings	29	2874	2903
3	Ground	5	474	479
4	Wood	91	9032	9123
5	Vineyard	105	10396	10501
6	Roads	32	3142	3174

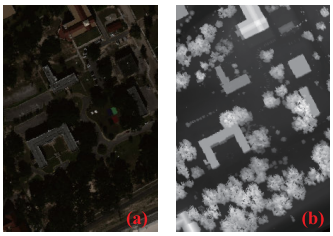


Fig. 4. Visualization of the MUUFL Dataset, including (a) HS pseudo color image and (b) LiDAR DSM gray image.

The former was recorded by the AISA Eagle sensor and has 63 channels ranging from 402.89 nm to 989.09 nm. The latter was recorded by the Optech ALTM 3100EA sensor. Their shapes and spatial resolutions are 166 × 600 and 1 m. The sample numbers in diverse categories are shown in Table I. Its visualization is shown in Fig. 3.

The Muufl Gulfport (MUUFL) dataset was acquired over the University of Southern Mississippi Gulf Park Campus, Mississippi, USA. Its HS data were recorded by the CASI-1500 sensor and has 64 channels ranging from 375 to 1050 nm. The shapes and spatial resolutions of its HS and LiDAR derived DSM data are 325 × 220 and 1 m, respectively. The sample numbers in different categories are shown in Table II. Its visualization is shown in Fig. 4.

The Houston dataset was acquired over the campus of Houston University by the NCALM sensor. Its HS data has 48 channels ranging from 380 to 1050 nm. The spatial resolutions of its HS and LiDAR derived DSM data are 1 m and 50 cm, respectively. Considering their shape difference, an interval sampling strategy was selected. One partitioned subscene with 601 × 596 samples (i.e., UH_NAD83_272652_3289689) was selected since the scale of the whole dataset is large. The sample numbers in different categories are listed in Table III. Its visualization is shown in Fig. 5.

In these three datasets, HS and LiDAR derived DSM data are chosen for joint classification. The categories with small sample ratios (i.e., < 2‰) were removed directly. The sample ratio is the ratio of the sample number in one category to

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TABLE II
DETAILED SAMPLE NUMBERS IN THE MUUFL DATASET

Class	Land-cover Category	Training	Testing	Total
1	Trees	232	23014	23246
2	Mostly Grass	42	4228	4270
3	Mixed Ground Surface	69	6813	6882
4	Dirt and Sand	18	1808	1826
5	Road	67	6620	6687
6	Water	5	461	466
7	Building Shadow	22	2211	2233
8	Building	62	6178	6240
9	Sidewalk	14	1371	1385
10	Yellow Curb	2	181	183
11	Cloth Panels	3	266	269

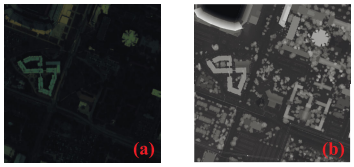


Fig. 5. Visualization of the Houston Dataset, including (a) HS pseudo color image and (b) LiDAR DSM gray image.

TABLE III
DETAILED SAMPLE NUMBERS IN THE HOUSTON DATASET

Class	Land-cover Category	Training	Testing	Total
1	Healthy Grass	14	1439	1453
2	Stressed Grass	51	5020	5071
3	Evergreen Trees	64	6371	6435
4	Deciduous Trees	30	2925	2955
5	Residential Buildings	81	7982	8063
6	Non-Residential Buildings	482	47782	48264
7	Roads	70	6969	7039
8	Sidewalks	125	12372	12497
9	Major Thoroughfares	179	17722	17901
10	Paved Parking Lots	42	4114	4156
11	Cars	11	1090	1101
12	Stadium Seats	68	6756	6824

the total sample number. For the optimization procedure, 1% samples in each category were randomly selected for training and the remaining 99% samples were used for testing.

B. Experimental Setting

The experiments were conducted on a PC (16.0-GB RAM) with Intel[1] Core[2] i7-10750H CPU and with the NVIDIA GeForce RTX 1650 GPU. The codes were carried out in the Keras framework. The adaptive moment estimation (Adam) was chosen as the network optimizer. The loss weights α, β, and γ were set as 0.05, 0.01, and 0.01, respectively. The learning rate was set as 5×10^{-3}. The batch size was selected within $\{128, 256, 512, 1024\}$. The scale number k was set as 3 and the scale combination $\{w_1, w_2, \ldots, w_k\}$ was selected within $\{3, 5, 7, 9\}$. The selection of these parameters is analyzed in Section IV-D. The experiments were replicated ten times to avoid random selection bias. The optimization procedures would proceed until the network parameters become stable. To evaluate the classification performance, five metrics were chosen [6], including the accuracy per category, overall accuracy (OA), average accuracy (AA), precision (PR), and $F1$-score ($F1$). Fourteen state-of-the-art techniques were chosen for comparison, including three baselines (i.e., early fusion, middle fusion, and late fusion) [10], ENDnet [10], two CNN-based networks (i.e., CNN_H and CNN_L) [37], enhanced multisource fusion network (EMFnet) [38], multiview hierarchical network (MVHN) [39], self-calibrated convolution-based multiscale DL network (MSNetSC) [40], local information interaction transformer (LIIT) [41], morphological convolution, and attention calibration network (MCAC) [42], $\text{HI}^2\text{D}^2\text{FNet}$ [6], Sal^2RN [7], and hashing-based deep metric learning (HDML) [43]. Their structures and experimental settings were fine-tuned for a fair comparison. Their descriptions are provided as follows.

1) *ENDnet* only uses FC layers to reconstruct its multimodal inputs, without considering any spatial information.
2) *CNN_H* and **CNN_L** classify HS and LiDAR samples, respectively, by three stacked CNN layers and one FC layer activated by the Softmax function.
3) *EMFnet* addresses feature extraction and modality fusion by one end-to-end network with feature tuning modules and one weighted summation strategy.
4) *MVHN* initially combines principal components analysis with Gabor filters for capturing cross-modal features and subsequently presents a residual network based stacking mechanism for HS and LiDAR feature fusion.
5) *MSNetSC* combines one hierarchical residual structure with self-calibrated convolution to enhance the feature representation ability. Several self-attention mechanisms are considered for multiscale features.
6) *LIIT* constructs one multihead self-attention guided local information interaction transformer for aligning HS and LiDAR information.
7) *MCAC* contains one morphological convolution and attention calibration network, which comprehensively extracts HS and LiDAR features for joint classification.
8) *$\text{HI}^2\text{D}^2\text{FNet}$* fuses horizontal geometry information in shading components with vertical geometry information in LiDAR data to increase the fusion interpretability.
9) *Sal^2RN* designs one cross-layer interaction module and one customized center spectrum correction module for LSS.
10) *HDML* extracts high-discrimination fused features by attention motivated feature fusion modules for single-source and cross-source feature fusion.

C. Experimental Results and Comparison

The classification accuracies are recorded in Tables IV–VI, respectively. The best accuracies per row are marked in bold. The pixelwise techniques, i.e., Early, Middle, Late, and ENDnet, cannot comprehensively capture land-cover descriptions in the spatial domain, making themselves hard to generate satisfactory performance. Notably, ENDnet works worse than these three baselines. These approaches were not selected on

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TABLE IV
CLASSIFICATION RESULTS, INCLUDING OA (%), AA (%), PR (%), AND F1 (×100), ON THE TRENTO DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	EMFnet	MVHN	MSNetSC	LIIT	MCAC	HI^2D^2FNet	Sal^2RN	HDML	Our
1	66.85	64.37	82.37	69.23	99.10	89.26	97.25	97.45	92.21	96.19	93.16	**99.62**	98.60	98.97	98.65
2	97.63	95.09	97.22	95.06	89.98	79.75	96.87	92.14	97.77	94.33	**99.10**	98.57	94.89	91.75	97.95
3	91.14	93.88	86.29	78.69	94.73	13.71	94.30	87.76	90.08	**98.10**	91.98	93.88	86.92	92.83	94.30
4	99.25	98.77	99.07	98.48	99.35	92.36	99.97	99.68	98.49	99.79	**100.00**	**100.00**	99.97	99.86	**100.00**
5	90.04	91.84	88.27	87.17	99.84	98.02	99.10	98.03	96.50	99.90	**100.00**	99.93	98.86	98.94	99.96
6	89.88	90.80	87.62	91.15	77.59	76.89	95.51	87.11	92.17	97.58	92.78	94.56	93.41	96.94	**97.96**
OA	90.46	90.50	91.51	89.23	96.23	89.83	98.45	96.57	96.10	98.57	98.11	99.12	98.01	98.22	**99.30**
AA	89.13	89.13	90.14	86.63	93.43	75.00	97.17	93.69	94.54	97.65	96.17	97.76	95.44	96.55	**98.14**
κ	87.22	87.24	88.71	85.63	94.96	86.33	97.93	95.43	94.80	98.09	97.48	98.83	97.35	97.63	**99.07**
PR	88.77	90.52	88.71	88.16	91.74	81.87	96.13	93.84	92.81	95.68	98.01	98.07	97.23	97.25	**99.11**
F1	88.85	89.72	89.26	87.36	92.31	76.40	96.61	93.74	93.60	96.52	97.03	97.90	96.25	96.86	**98.61**

TABLE V
CLASSIFICATION RESULTS, INCLUDING OA (%), AA (%), PR (%), AND $F1$ (×100), ON THE MUUFL DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	EMFnet	MVHN	MSNetSC	LIIT	MCAC	HI^2D^2FNet	Sal^2RN	HDML	Our
1	94.42	94.36	92.64	93.75	96.57	84.27	96.90	96.58	96.20	96.72	**97.74**	96.95	95.28	97.21	96.13
2	51.40	64.10	51.49	71.14	63.55	33.96	71.81	66.75	74.01	66.91	69.70	77.51	73.89	69.37	**87.80**
3	82.03	65.60	74.08	66.74	76.12	35.40	81.56	73.36	81.08	79.14	83.21	**84.15**	80.41	77.78	80.99
4	69.03	73.34	80.92	47.51	59.02	4.48	74.61	56.80	**91.54**	75.88	79.76	89.88	88.44	77.71	77.49
5	93.23	86.72	91.10	88.94	93.04	62.81	91.50	83.08	92.66	91.47	93.69	92.60	**94.64**	90.95	93.41
6	72.89	83.73	73.97	82.00	45.55	74.40	81.56	41.00	95.66	89.15	59.22	**96.10**	85.03	63.12	75.49
7	47.04	64.18	62.96	67.62	62.60	15.65	75.03	65.08	81.28	69.74	70.87	**83.49**	77.02	75.35	75.49
8	86.73	90.84	88.94	91.19	95.03	76.17	95.39	93.40	**97.17**	95.60	95.00	95.95	95.40	96.62	96.71
9	57.91	50.40	**61.05**	47.78	41.58	2.41	34.65	17.29	34.21	32.53	45.15	31.00	33.77	45.73	56.60
10	24.86	**28.73**	25.41	15.47	0	0.55	3.31	2.21	0	25.41	7.18	1.10	0.55	3.87	11.05
11	41.73	59.40	56.39	**87.97**	66.92	0	60.90	42.11	40.98	45.11	45.11	50.00	59.02	31.58	63.53
OA	83.90	83.31	83.35	83.35	85.67	61.92	88.18	83.40	89.18	87.26	88.83	90.03	88.43	87.82	**90.11**
AA	65.57	69.22	69.00	69.10	63.63	35.46	69.75	57.97	71.34	69.79	67.88	72.61	71.22	66.30	**74.06**
κ	78.60	77.97	78.04	78.00	80.92	49.11	84.29	77.86	85.68	83.08	85.10	86.80	84.68	83.80	**86.95**
PR	75.19	68.96	73.87	68.61	72.55	37.56	76.22	65.35	71.65	72.95	77.45	77.45	77.88	**81.11**	78.41
F1	68.40	68.43	70.55	68.71	67.02	35.21	72.13	60.62	72.50	70.75	71.62	74.09	73.23	69.83	**75.79**

TABLE VI
CLASSIFICATION RESULTS, INCLUDING OA (%), AA (%), PR (%), AND $F1$ (×100), ON THE HOUSTON DATASET

Class	EMFnet	MVHN	MSNetSC	LIIT	MCAC	HI^2D^2FNet	Sal^2RN	HDML	Our
1	78.74	75.82	78.39	50.73	60.81	74.70	71.16	**83.60**	78.32
2	79.48	83.17	77.03	74.96	78.41	78.55	89.44	83.82	**90.50**
3	87.47	75.53	85.73	86.38	89.70	86.89	95.21	96.88	**97.19**
4	73.23	78.36	82.43	84.68	62.19	86.97	83.86	87.69	**89.91**
5	75.28	79.27	76.35	**95.16**	85.40	88.51	86.22	91.73	93.92
6	96.57	96.17	93.27	93.89	95.02	93.42	96.81	97.01	**97.25**
7	32.97	38.79	36.65	44.08	36.05	36.10	46.85	44.64	**57.24**
8	71.57	65.71	68.63	58.97	60.04	70.44	56.98	71.86	**74.57**
9	79.35	76.83	72.23	78.15	82.50	82.46	89.74	86.60	**92.28**
10	85.51	87.68	86.63	89.57	83.13	83.01	89.67	91.61	**94.26**
11	**67.89**	57.52	29.27	54.40	59.54	57.34	51.38	28.26	55.41
12	96.60	93.18	94.46	96.05	97.57	95.54	98.08	96.48	**98.79**
OA	83.78	82.65	81.00	82.89	82.78	83.91	86.50	87.74	**90.58**
AA	77.06	75.67	73.42	75.59	74.20	77.83	79.62	80.02	**84.97**
κ	79.38	78.00	76.05	78.48	78.26	79.78	82.90	84.51	**88.12**
PR	79.09	75.89	75.03	78.54	77.28	78.83	81.22	85.24	**87.29**
F1	77.81	75.42	73.15	76.46	75.43	78.00	80.03	81.16	**85.69**

the Houston dataset for simplification. In terms of spatial descriptions, CNN_H and CNN_L perform worse than other cross-modal techniques since these two methods only consider one RS modality. These results prove that fusing both RS modalities is capable of obtaining comprehensive information from complementary information sources. In these patchwise techniques, MAMInet II increases the OA by 0.18%, 0.08%, and 2.84% than the best approach on these three datasets, respectively. For the MUUFL dataset, although the PR of HDML is larger than that of MAMInet II, all accuracies on these datasets collectively illustrate the superiority of MAMInet II in terms of structure and strategy. The accuracies of LIIT and MCAC with attention mechanisms are worse than those of MAMInet II, showing that considering HIID in HS and LiDAR joint classification is effective. HI^2D^2FNet works worse than MAMInet II, further verifying the competitive property of the presented MAI capturing strategy. The tenth category (its sample ratio is 2.56‰) in the MUUFL dataset has relatively low accuracies since there are 183 *Yellow Curb* samples in total. Thus, only two samples were considered during training procedures. In this difficult circumstance, MAMInet II performs better than other patchwise techniques.

The classification maps are provided in Figs. 6–8, respectively. Apparently, MAMInet II is capable of generating more accurate classification maps with fewer wrongly classified samples than other comparison techniques. This conclusion is consistent with those drawn from the quantitative accuracies. Moreover, MAMInet II performs better in classifying the samples in small and crowded regions. For example, on the Houston dataset, the *Stressed Grass* and *Evergreen Trees* samples can be better classified by the proposed techniques although they are surrounded by disruptive samples. Furthermore, the *Wood* samples in the Trento dataset and the *Mostly Grass* samples in the MUUFL dataset can be accurately identified by MAMInet II. These maps and accuracies

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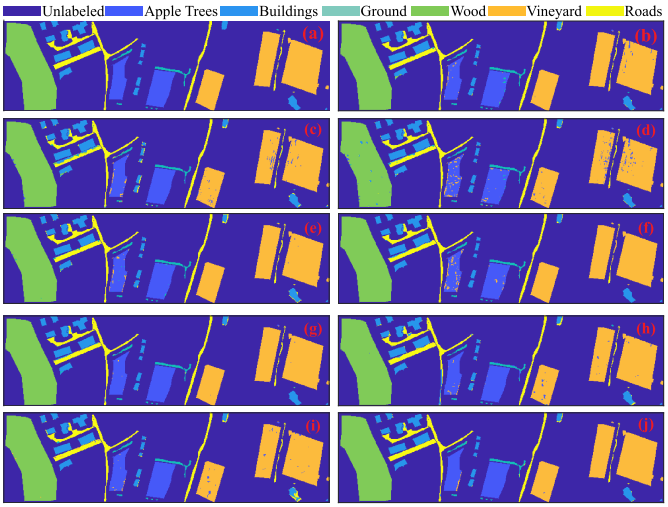


Fig. 6. (a) Ground truth and (b)–(j) classification maps on the Trento dataset, including (b) EMFnet, (c) MVHN, (d) MSNetSC, (e) LIIT, (f) MCAC, (g) HI^{2}D^{2}FNet, (h) Sal^{2}RN, (i) HDML, and (j) MAMInet II.

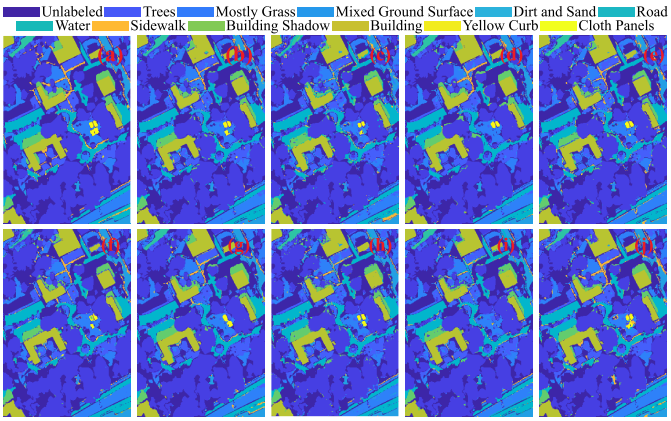


Fig. 7. (a) Ground truth and (b)–(j) classification maps on the MUUFL dataset, including (b) EMFnet, (c) MVHN, (d) MSNetSC, (e) LIIT, (f) MCAC, (g) HI^{2}D^{2}FNet, (h) Sal^{2}RN, (i) HDML, and (j) MAMInet II.

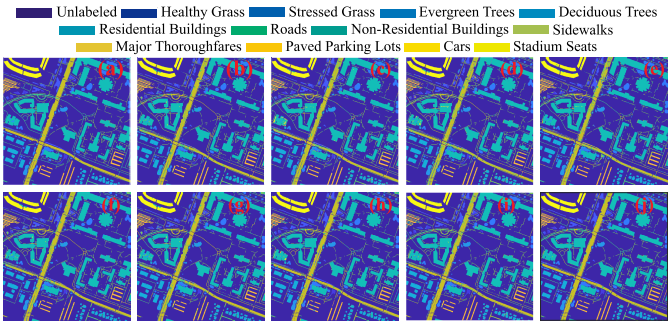


Fig. 8. (a) Ground truth and (b)–(j) classification maps on the Houston dataset, including (b) EMFnet, (c) MVHN, (d) MSNetSC, (e) LIIT, (f) MCAC, (g) HI^{2}D^{2}FNet, (h) Sal^{2}RN, (i) HDML, and (j) MAMInet II.

collectively illustrate the effectiveness of MAMInet II qualitatively and quantitatively.

Fig. 9 shows the visualization of the decomposed reflectance and shading components by MAMInet II to prove the HIID performance and fusion interpretability. It proves that the generated reflectance component reflects object materials in uniform surrounding illumination. On the contrary, the shading component describes detailed illumination rather than object properties. Some typical regions are squared. In the Trento dataset, the observed area has a comparatively small elevation difference. Although the local sample gradients in its components are small, the nonuniform illumination in (S-1) is captured. The squared buildings in the MUUFL and Houston datasets further illustrate this decomposition interpretability regarding semantic information. In MAMInet II, by removing the shading component, the sample consistency within the same regions is enhanced, providing reliable discriminant information for joint classification.

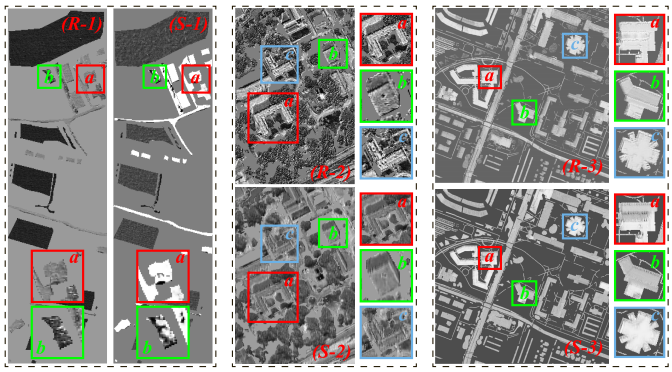


Fig. 9. Decomposed components by MAMInet II, (R-*) and (S-*) correspond to the reflectance and shading components, (*-1)-(*-3) correspond to the three datasets, typical regions are squared and enlarged.

D. Ablation Study and Parameter Selection

Multiple ablation experiments were conducted on the Houston dataset to prove the effectiveness of MAMInet II.

1) Necessity of Both RS Modalities: Several scenarios were provided to prove the necessity of both RS modalities as follows.

1) *HS+∅:* Using the HS data as the MAMInet II input and removing the LiDAR-related structure.
2) *LiDAR+∅:* Using the LiDAR data as the MAMInet II input and removing the HS-related structure.
3) *HS+HS:* The two inputs are both the HS data.
4) *LiDAR+LiDAR:* The two inputs are both the LiDAR data.
5) *HS+LiDAR (the Presented Structure):* The two inputs of MAMInet II are the HS and LiDAR data, respectively.

The classification results are provided in Table VII. It proves that only using the LiDAR data cannot yield satisfactory classification results since the object spectral properties are not available in this RS modality. When using one single RS modality, all metrics are lower than those of the proposed structure. Specifically, HS+LiDAR increases the OA by 4.94% and 26.56% than HS+∅ and LiDAR+∅, respectively. Extracting features from two identical RS modalities does not contribute to increasing the accuracies. These results clearly verify that considering HS and LiDAR data in accurately identifying on-ground objects is feasible and necessary.

2) Necessity of Network Structure: To verify the necessity of the presented modules and mechanisms in MAMInet II, several alternative scenarios are provided as follows.

1) *LSS-Related Scenarios:*
   a) $\boldsymbol{S_1}$*:* Removes the LSS module from MAMInet II.

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TABLE VII

CLASSIFICATION RESULTS ON THE HOUSTON DATASET TO PROVE THE NECESSITY OF BOTH RS MODALITIES

Scenarios	OA (%)	AA (%)	PR (%)	F1 (×100)
HS+∅	85.64	77.01	82.45	78.23
LiDAR+∅	64.02	31.65	36.01	29.59
HS+HS	86.35	77.36	82.25	79.15
LiDAR+LiDAR	61.72	27.73	33.21	25.23
HS+LiDAR	**90.58**	**84.97**	**87.29**	**85.69**

TABLE VIII

CLASSIFICATION RESULTS, INCLUDING OA (%), AA (%), PR (%), AND $F1$ (×100), ON THE HOUSTON DATASET TO PROVE THE NECESSITY OF NETWORK STRUCTURE

Metric	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	Our
OA	85.59	85.01	87.86	86.18	82.20	88.09	87.71	88.60	**90.58**
AA	75.56	74.04	80.99	79.22	65.92	82.38	82.93	83.35	**84.97**
PR	81.98	81.45	85.41	81.30	72.70	84.11	83.30	84.30	**87.29**
F1	76.40	74.45	82.55	79.86	66.79	82.75	82.93	83.70	**85.69**

2) *IIR-Related Scenarios:*
   a) $\boldsymbol{S_2}$*:* Removes the IIR module from MAMInet II.
   b) $\boldsymbol{S_3}$*:* Preserves the shading component and removes the reflectance component for the GDS module.
   c) $\boldsymbol{S_4}$*:* Concatenates the reflectance and shading components for the following GDS module.
3) *GDS-Related Scenarios.*
   a) $\boldsymbol{S_5}$*:* Removes the GDS module from MAMInet II.
   b) $\boldsymbol{S_6}$*:* Replaces the concatenation operator with the addition operator for generating the MAI features.
   c) $\boldsymbol{S_7}$*:* Removes the generative switching mechanism and replaces the VAE based structure with an AE based one.
   d) $\boldsymbol{S_8}$*:* Removes the generative switching mechanism and preserves the VAE based structure.

The classification accuracies of these alternative scenarios are listed in Table VIII. Initially, S_1 removes the LSS module, meaning that the raw RS samples rather than their high-level features are delivered into the IIR module. Its performance is unsatisfactory since its OA is 4.99% lower than that of MAMInet II. Subsequently, the effectiveness of the IIR module is fully verified. The results of S_2 show that the contribution of this module cannot be ignored since it simulates nonuniform illumination by considering the HS and LiDAR geological connection. S_3 and S_4 use the shading component and yield unsatisfactory performance since this component does not reflect detailed material information and records unnecessary surrounding illumination. Although the performance of S_3 is slightly higher than that of S_4, it does not mean that the shading component contains more discriminant information than the reflection component. Apparently, the properties of both components are absolutely diverse, making that concatenating them directly cannot achieve a satisfactory balance. The concatenated components tend to optimize the whole network toward two different directions, further interfering with the joint classification process. S_5–S_8 illustrates the contribution of the GDS module. S_5 removes the GDS module directly and decreases the OA by 8.38%. The accuracies of S_6 show that the concatenation operator is capable of balancing and enhancing the contributions of the MAI features $\{\boldsymbol{M_1}, \boldsymbol{M_2}, \ldots, \boldsymbol{M_k}\}$ in several scales. Although the addition operator can also fuse these MAI features, it tends to weaken their unique properties. S_7 and S_8 both remove the generative switching mechanism and generate worse results than MAMInet II. This unique mechanism contributes to closing the cross-modal RS distribution gap. Furthermore, S_7 replaces the VAE-based structure with an AE-based one. Compared with S_8, S_7 decreases these metrics by 0.89%, 0.42%, 1.00%, and 0.77%. The reason is that the utilized structure provides opportunities for cross-modal interaction and outperforms other structures in terms of the information mining potential.

TABLE IX

SCENARIO EXPLANATIONS AND CLASSIFICATION RESULTS, INCLUDING OA (%), AA (%), PR (%), AND $F1$ (×100), ON THE HOUSTON DATASET TO PROVE THE NECESSITY OF LOSS ELEMENTS

Loss	K_1	K_2	K_3	K_4	K_5	K_6	K_7	K_8	K_9	Our
L_D	×	✓	✓	✓	✓	×	✓	✓	×	✓
L_M	✓	×	✓	✓	✓	×	✓	✓	×	✓
L_R	✓	✓	×	✓	×	×	✓	✓	×	✓
L_S	✓	✓	✓	×	×	×	✓	✓	×	✓
L_X	✓	✓	✓	✓	✓	✓	✓	×	✓	✓
L_{KL}	✓	✓	✓	✓	✓	✓	×	×	×	✓
L_C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OA	88.48	89.43	88.52	85.84	83.94	89.17	88.30	89.35	89.57	**90.58**
AA	82.81	83.32	82.88	79.62	75.61	82.61	82.92	82.91	**84.99**	84.97
PR	84.99	86.41	83.94	80.55	78.73	85.79	84.48	86.96	87.16	**87.29**
F1	83.44	84.33	83.07	79.37	75.95	83.68	83.39	84.46	**85.73**	85.69

3) Necessity of Loss Elements: The contributions of the loss elements in (13) are analyzed by presenting several alternative scenarios, including $\{K_1, K_2, \ldots, K_9\}$. These scenarios removing some loss elements are explained in Table IX. The accuracies are also recorded. Apparently, the removal of any loss elements tends to decrease these metrics comprehensively. The reason is that this removal tends to weaken the constraints on the whole optimization procedure. The results further illustrate that worse appearances can be generated when preserving the decomposition structures and removing the decomposition-related loss elements (i.e., L_R and L_S). K_9 removes the HIID-related loss elements (i.e., L_D, L_R and L_S), L_M and L_{KL}. Thus, the HIID- and VAE-related structures degenerate into classic feature extraction and AE-related structures, respectively. Although the constraints on these structures are removed, these structures are effective enough. The OA and PR of K_9 are decreased by 1.01% and 0.13% than those of MAMInet II, further verifying the feasibility of the presented network structure. Moreover, when removing the aforementioned structures, the strong fusion interpretability of MAMInet II disappears, reducing the application potential. However, the AA and $F1$ of K_9 are slightly larger than those of MAMInet II. The reason is that the VAE-based structure without corresponding loss elements interferes with the updating process, making the optimization direction inconsistent with the classification goal.

4) Network Scale Selection: The optimal scale number k and scale combination $\{w_1, w_2, \ldots, w_k\}$ were selected within $\{1, 2, 3, 4\}$ and $\{3, 5, 7, 9\}$, respectively. The results in terms of four metrics and the training time per epoch are listed in Table X. The results clearly show that using multiple scales contributes to yielding a better joint appearance. Using one single scale in MAMInet II meets challenges in capturing comprehensive discriminant information. Furthermore, the

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TABLE X

CLASSIFICATION RESULTS AND TRAINING TIME PER EPOCH WITH DIFFERENT SCALE COMBINATIONS ON THE HOUSTON DATASET TO SELECT OPTIMAL SCALES FOR MAMINET II

Scale Combination	OA (%)	AA (%)	PR (%)	F1 (×100)	Time (s)
{3}	83.96	75.97	77.23	76.38	**0.48**
{5}	88.23	81.88	85.35	82.93	0.88
{7}	89.31	84.36	85.54	84.64	0.94
{9}	89.52	84.10	85.91	84.50	1.17
{3, 5}	87.01	80.62	82.49	81.18	1.32
{3, 7}	88.81	83.94	85.14	84.25	1.92
{3, 9}	89.61	84.39	85.92	84.99	2.04
{5, 7}	88.99	82.13	86.58	83.40	1.96
{5, 9}	89.90	84.55	87.03	85.59	2.45
{7, 9}	89.76	84.75	85.97	85.05	2.88
{3, 5, 7}	89.41	84.69	85.24	84.77	3.23
{3, 5, 9}	**90.58**	**84.97**	**87.29**	**85.69**	4.23
{3, 7, 9}	89.98	84.88	86.80	85.57	4.42
{5, 7, 9}	89.37	84.28	85.93	84.99	4.46
{3, 5, 7, 9}	89.02	82.752	84.90	83.50	4.92

TABLE XI

CLASSIFICATION RESULTS ON THE HOUSTON DATASET TO SELECT OPTIMAL LOSS WEIGHTS WHEN THE VALUES OF α AND β ARE CONSISTENT

α and β	OA (%)	AA (%)	PR (%)	F1 (×100)
1	76.93	55.20	70.06	52.95
0.05	90.05	**85.86**	86.34	85.96
0.01	**90.11**	84.05	**87.58**	85.38
0.005	88.63	83.75	85.10	**86.11**
0.001	90.03	85.50	86.83	84.18

scale number growth generally corresponds to the increase in the training time. Larger scales generally correspond to longer training time and lower training efficiency. The average training time per epoch is increased by 3.22 s when k changes from 1 to 3. When k changes from 3 to 4, the average OA is decreased by 0.82%. However, the average training time per epoch is increased by 0.84 s, making the whole procedure inefficient. Considering both the classification performance and efficiency, $\{3, 5, 9\}$ ($k = 3$) was regarded as the optimal scale combination and selected in MAMInet II.

E. Loss Weight Selection

Experiments were conducted on the Houston dataset to select the optimal loss weights, i.e., α and β. These weights measure the contributions of the corresponding loss elements. Two separate scenarios are considered since the values of α and β can be consistent and inconsistent generally. Specifically, α and β were selected within $\{0.001, 0.005, 0.01, 0.05, 1\}$ when their values are consistent. When their values are inconsistent, they were selected within $\{0.005, 0.05, 0.5, 5\}$ and $\{0.001, 0.01, 0.1, 1, 10\}$, respectively. The results are shown in Table XI and Fig. 10. Notably, small loss weights do not always correspond to superfluous and unnecessary loss elements. They are always determined by the orders of magnitude. Specifically, the loss elements corresponding to α are generally based on HIID and KL divergence, and the loss elements corresponding to β are based on the VAE structure. These loss elements are always hard to minimize, making their values larger than the L_C values. If these weights vary slightly, the contribution of L_C will be ignored during the training procedure, weakening the final

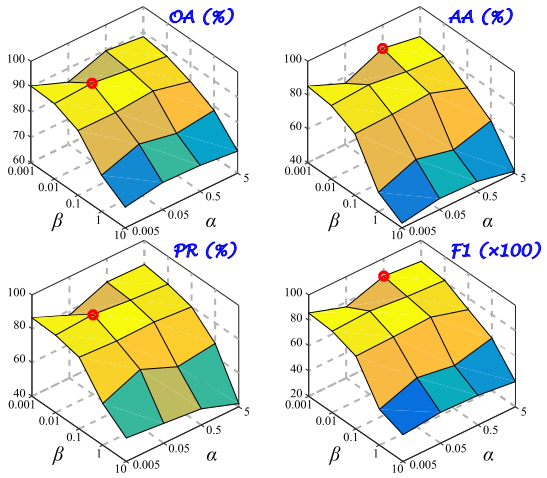


Fig. 10. Classification results on the Houston dataset to select optimal loss weights when the values of α and β are inconsistent. The best results are highlighted by red circles.

performance. It shows that the structures with consistent loss weights generally perform worse than those with inconsistent loss weights. Notably, the consistent value couples in Table XI are not selected since their performances are unsatisfactory. In Fig. 10, the best result appears when $\alpha = 0.05$ and $\beta = 0.01$. These values were chosen as the default weights.

V. CONCLUSION

This article has presented a novel HS and LiDAR joint classification method called MAMInet II to strengthen the land-cover classification capability using both RS modalities simultaneously in real-world scenarios with nonuniform surrounding illumination. MAMInet II contributes to solving two primary challenges, including illumination interference and heterogeneous simulation. Our motivation is to eliminate CII and capture high-level MAI from the complementary HS and LiDAR data in terms of their geological connection. MAMInet II simulates real-world illumination conditions by associating HS spectral curves with LiDAR elevations, increasing fusion capability and interoperability. The generative switching mechanism in MAMInet II further closes the cross-modal RS distribution gap and contributes to cross-modal interaction. The multiscale and multistage versions align local and global RS properties for simulation integrity. Experiments on three publicly available datasets verify the effectiveness of MAMInet II against 14 state-of-the-art techniques qualitatively and quantitatively. Multiple ablation studies and parameter selection experiments have been conducted.

Further experiments will focus on broadening the scope of applications of multimodal classification in RS-related regions.

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